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DAILYKHATA

Deliverable 3: Requirements Specification

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1. Introduction

This deliverable specifies the business, functional and non-functional requirements for DailyKhata. It develops the user needs in D1 and the requirement candidates in D2 into a manageable catalogue for design, implementation and evaluation. The catalogue contains eight business requirements, 33 functional requirements and 18 non-functional requirements. It describes intended behaviour, not completed implementation or successful testing. [1, 2]

DailyKhata is a personal expense management application primarily for Pakistani users. The primary users are homemakers and household budget managers, university students and salaried professionals. Small shopkeepers are secondary direct users for petty cash and supplier expenses. Family members and advisers benefit from information shared by the account holder. D1 contains the five agreed persona profiles; this specification uses them without redefining their identities or ages.

One account holder manages personal or household expenses. The application is mobile-first, uses PKR and targets ordinary Android-class devices. Its core joins reviewed receipt and cash capture with a private ledger, structured accounting records, product price history, summaries and a specialised, read-only AI Financial Assistant. Optional features retain their separate priorities.

The statement "shall" specifies behaviour within the priority shown. Must Have requirements define the minimum successful FYP. Should Have requirements are intended but deferrable; Could Have requirements are optional if time permits. A feature does not become mandatory merely because its statement uses "shall". Future possibilities and current exclusions are listed separately.

2. Requirements Sources and Approach

The main sources are D1's audience and problem analysis, D2's reported evidence and requirement candidates, and the agreed project scope. The official D3 assignment provides the distinction between business outcomes, system behaviour and quality requirements, together with the expectation of source links and validation. [1-4]

D2 reports three interviews, informal observation and a historical examination of existing applications. Those findings support convenient capture, confirmation, retrospective entry and spending visibility. The internal Chart of Accounts, semantic product classification, authentication and assistant capabilities also derive from project design decisions. They are not presented as additional interview findings. No new survey, interview or user trial is claimed in this deliverable.

Related behaviours are consolidated into one requirement where they form a logical capability. Receipt-field extraction is specified together, for example, while review remains separate because it establishes the user's control over saving. Shared data rules appear once below instead of being repeated beside every requirement. The D2 candidate identifiers provide a single traceability table in Section 7.

The essential workflow is: authenticate, capture or enter an expense, review its values and categories, reconcile its total, save it, and retrieve or analyse it. An unconfirmed draft never contributes to financial totals. Each saved expense has an owner, stable identity, expense date, creation timestamp, PKR amount and valid category/account allocations. Product attributes are retained where available. All protected operations use the same account boundary.

Requirements are written to allow observable checks without prescribing a particular framework, database or model provider. Numerical quality targets are retained from the detailed specification as proposals for pilot and supervisor review. They are not results of the reported D2 research. The assignment's illustrative hospital requirements, uptime and concurrency examples are not DailyKhata requirements.

3. Business Requirements

Business requirements describe the outcomes sought by stakeholders. Their achievement depends on the related functions and quality conditions, rather than on unmeasured promises of financial savings or user retention.

ID	Business requirement	Priority
BR-01	Enable users to maintain expense records with less repetitive entry, whether a purchase has a receipt or not.	Must Have
BR-02	Provide trustworthy, correctable expense histories whose classifications, totals and accounting records remain coherent.	Must Have
BR-03	Help account holders understand recorded spending, explain changes and identify possible savings from their own history.	Must Have
BR-04	Support informed historical product-price comparisons while retaining differences in brand, product and quantity.	Must Have
BR-05	Protect personal financial information through account-specific access and appropriate data handling.	Must Have
BR-06	Make core expense workflows understandable and usable on ordinary Android-class devices, including recovery from service failures.	Must Have
BR-07	Improve presentation and voluntary sharing through optional language, label, view, image and export features.	Should Have
BR-08	Support additional recording habits through optional personal categories, offline drafts, reminders and household summaries.	Could Have

The first six outcomes define the minimum product purpose. A useful demonstration should show that a user can save a corrected receipt, record a cash purchase, retrieve both, and obtain consistent summaries and answers. Product comparisons should retain their

purchase context, and another account must not gain access to those records. Optional outcomes improve convenience without becoming dependencies of the core workflow.

4. Functional Requirements

4.1 Authentication and Profile

Authentication establishes the ownership boundary for expenses, receipt information, product histories and assistant queries. Basic preferences are deliberately limited so that profile management does not introduce household roles or organisation administration.

ID	Requirement	Priority
FR-AUTH-01	The system shall register a distinct account from valid unused credentials, establish a session after successful verification, and terminate access when the user logs out. Invalid input shall create no partial account.	Must Have
FR-AUTH-02	The system shall authorise every protected read, change, deletion, export and assistant retrieval against the current account, rejecting unauthenticated or cross-account requests without disclosing protected data.	Must Have
FR-AUTH-03	The system shall let the account holder view and update a display name and retain a default summary month between authenticated sessions without changing account identity or record ownership.	Must Have

The current month is the initial summary default. Failed profile updates preserve the previous value. Logout ends local access immediately; server session handling must follow the session policy even if connectivity is unavailable. Invalid login messages should not disclose whether a particular account exists.

4.2 Expense Capture

Both core no-receipt routes produce a draft for the same review and saving process. They require no receipt image and support retrospective recording. Expense dates are separate from creation timestamps, and relative dates such as "yesterday" are resolved using Asia/Karachi and shown for confirmation.

ID	Requirement	Priority
FR-CAP-01	The system shall create a reviewable manual expense draft from an amount, expense date and short description, allowing a past purchase date independently of its creation time.	Must Have
FR-CAP-02	The system shall interpret a short text-based cash-expense description into proposed fields and allow the user to review and correct them before saving. Missing or ambiguous values shall require clarification.	Must Have

A visible current-date default is acceptable for manual entry. It must not silently replace a missing or ambiguous receipt date. Expense amounts are positive PKR values represented to two decimal places. Refunds, income entry and scheduled future expenses are not introduced by these capture routes.

4.3 Receipt OCR and Review

OCR converts an image into text; structured extraction interprets that text into an expense draft. Neither operation confirms a financial record. Original OCR text remains separate from corrected and inferred values while the record exists.

ID	Requirement	Priority
FR-OCR-01	The system shall accept a camera photograph or supported uploaded receipt image, process it using OCR, and preserve the original recognised text separately from corrections. It shall show processing state and permit user-triggered retry after recoverable failure.	Must Have
FR-OCR-02	The system shall propose structured receipt data including merchant and date where available, purchase line items, available quantities and units, unit prices, line totals and payable total. Unknown or ambiguous values shall remain identifiable for review.	Must Have
FR-OCR-03	The system shall present extracted fields and uncertainty for review, allow correction of all extracted expense fields, and let the user discard the draft or complete it manually before explicit confirmation.	Must Have

Camera-permission denial should leave upload or manual entry available. Rejected images should explain the accepted format and size. A missing merchant does not prevent manual fallback; a missing or conflicting payable total must be resolved before saving. An unknown quantity is not silently treated as one. Purchased count, pack size, unit price and line total remain distinct.

Core draft recovery is limited to the current open authenticated session. Persistent offline drafts are optional. Receipt image retention beyond processing is also optional, so successful structured extraction must not depend on permanent image archiving.

4.4 Categorisation and Accounting

User-facing categories and internal accounts serve different purposes. Categories provide familiar spending labels, while the Chart of Accounts supplies stable accounting references. A category name change must not silently change its meaning or account mapping.

ID	Requirement	Priority
FR-CAT-01	The system shall propose categories for each transaction and each receipt line item, present locally understandable labels, and retain the user's confirmation or corrections with the saved expense.	Must Have
FR-CAT-02	The system shall maintain a full internal Chart of Accounts with stable identifiers, names, account types and active status, and map every saved expense allocation through its category to a valid account.	Must Have
FR-CAT-03	The system shall store accounting-compatible expenses whose reviewed allocations equal the payable total, reject unresolved differences, and enforce equal debits and credits for every committed journal wherever double-entry is implemented.	Must Have

The account framework includes asset, liability, equity, income and expense types, but exposes only expense-management workflows. Cash is an asset account. Recognising an income account type does not introduce a sales or income-entry workflow. Final account codes and hierarchy remain design choices. Users should not need to enter journals or account codes.

For a mixed receipt, category totals use the allocated item amounts rather than counting both the receipt total and its items. An unitemised expense has a single allocation equal to its total. Discounts, tax and rounding adjustments require explicit user-reviewed allocation; they remain separate from products and do not enter price history. The primary transaction category supports display and search without replacing the mixed-item allocations.

Double-entry is not made unconditional by this draft. If it is selected, every journal, including corrections, must balance. Otherwise, the implementation must still maintain valid account mappings and reconciled expense allocations. This preserves structured records without claiming complete SME accounting or accounting certification.

4.5 Product Classification and Price History

Semantic classification groups related products for analysis without making them identical. A product type such as Laundry Detergent may include different brands and pack sizes. Exact product histories must retain those differences.

ID	Requirement	Priority
FR-PROD-01	The system shall keep original receipt text, corrected item name, individual product identity, brand, canonical product type, expense category and accounting classification distinct. It shall propose reviewable product types and allow correction of inferred attributes without merging distinct products.	Must Have

ID	Requirement	Priority
FR-PROD-02	The system shall display recorded product prices with expense dates, available quantities and links to their source transactions, identifying unresolved product matches rather than silently including them in an exact-product history.	Must Have
FR-PROD-03	The system shall compare related product prices using equivalent quantities and compatible units where known, show the comparison basis, and otherwise display recorded prices with an explicit limitation.	Must Have

For example, Surf Excel 1kg and Bonus 1kg can belong to the same product type while remaining separate products. Correcting a brand or type must not rewrite unrelated purchases. Unknown brands, quantities or identities remain unknown until resolved. A price comparison describes recorded purchases at their recorded dates; it does not establish current retailer prices or equal product quality.

4.6 Ledger, Search and Summaries

The ledger is the common source for current spending views, price history and assistant retrieval. Drafts and deleted expenses are excluded. A successful change must be reflected consistently before the next fresh query returns.

ID	Requirement	Priority
FR-LED-01	The system shall save one complete expense only after valid values and categories are explicitly confirmed, and display its detail and active history in descending expense-date order with a stable order for ties.	Must Have
FR-LED-02	The system shall search and filter the account holder's expenses by combined date range, category and text criteria, including relevant merchant or item text and mixed-receipt category allocations.	Must Have
FR-LED-03	The system shall support validated, user-confirmed editing and deletion of saved expenses, updating dependent accounting records, summaries and price histories consistently and detecting stale conflicting edits.	Must Have
FR-LED-04	The system shall provide category totals, calendar-month spending totals and transaction counts, plus a dashboard showing the selected month's total, category summary and recent expenses, without double counting.	Must Have

Empty history should offer a clear entry route. No search matches, zero recorded spending and a loading failure must be distinguishable. If a view shows an older snapshot

after a failure, it must be labelled. Deletion requires user confirmation; cancellation changes nothing, and a failed deletion must not be reported as successful.

Category totals use the allocation rule in Section 4.4. Monthly totals use expense dates, not record-creation dates. A receipt entered this month for a purchase last month belongs to the purchase month. This same interpretation applies to dashboard and assistant results.

4.7 AI Financial Assistant

The assistant specialises in the current user's DailyKhata financial records. Its access is read-only and account-specific. It must support the following capabilities as part of the core product, independently of optional comparison screens or visualisations.

ID	Requirement	Priority
FR-AI-01	The assistant shall accept natural-language spending questions, retain context for follow-up questions, clarify ambiguous categories or periods, and retrieve only the current account's authorised active records.	Must Have
FR-AI-02	The assistant shall answer transaction lookups, spending summaries, category and period comparisons, recurring-spending analysis and questions about unusually high recorded spending using supporting dates, amounts and an explicit comparison basis.	Must Have
FR-AI-03	The assistant shall support semantic product and historical price comparisons and personalised saving suggestions based on recorded history, retaining product differences, quantity limitations and the assumptions behind suggestions.	Must Have
FR-AI-04	The assistant shall distinguish recorded facts, calculations and suggestions; identify relevant periods or supporting records; explain missing data or unavailable services; and remain read-only, declining investment, tax, banking and unrelated financial-education requests.	Must Have

The assistant is not an unrestricted general financial chatbot. It cannot edit expenses, move money, execute payments or control external accounts. A lack of matching records means no recorded match, not proof that the purchase never happened. A zero comparison baseline cannot support an ordinary percentage-increase calculation, and incomplete periods must be identified.

Recurring analysis concerns repeated past expenses, not automatic future entries. Identifying comparatively high recorded spending is descriptive analysis, not predictive anomaly detection. Saving suggestions remain hypothetical choices based on available history. The system must not turn a calculated opportunity into a guaranteed saving or infer an unsupported cause for a change.

4.8 Optional Features

Optional features use the same ownership, correction and integrity rules when implemented. They remain independently deferrable; grouping them in this section does not require all of them to be selected together.

ID	Requirement	Priority
FR-OPT-01	The system shall support user-initiated voice expense capture through an editable transcript and the same reviewable draft as text entry.	Should Have
FR-OPT-02	The system shall offer English and Urdu interface selection, appropriate right-to-left Urdu presentation, and Urdu questions and answers for core assistant query types when bilingual support is implemented.	Should Have
FR-OPT-03	The system shall export the account holder's selected expenses in at least one of CSV or PDF, preserving their recorded values and scope.	Should Have
FR-OPT-04	The system shall allow custom displayed labels for existing categories without changing category identity or account mappings.	Should Have
FR-OPT-05	The system shall provide simple labelled spending visualisations with equivalent readable values that agree with category totals.	Should Have
FR-OPT-06	The system shall provide dedicated comparisons of two selected months, including absolute differences and appropriate handling of a zero baseline.	Should Have
FR-OPT-07	The system shall retain and retrieve receipt images linked to saved expenses, under account-specific access and the retention policy.	Should Have
FR-OPT-08	The system shall let users create personal expense categories mapped to existing appropriate expense accounts.	Could Have
FR-OPT-09	The system shall retain manual or text expense drafts locally while offline for later review and saving, without counting drafts as expenses or creating duplicate records on retry.	Could Have
FR-OPT-10	The system shall let users enable or disable a local reminder to record expenses.	Could Have
FR-OPT-11	The system shall export an account-holder-selected household expense summary without introducing shared accounts or household permissions.	Could Have

CSV and/or PDF means at least one format, not a commitment to both. If CSV is selected, fields must be escaped and formula-like user text exported as inert text. General

export can exist without a dedicated household summary. Similarly, custom labels rename existing categories; creating new category identities is a separate Could Have feature.

Urdu localisation is not guaranteed until implemented and reviewed. It includes mixed-direction readability for numbers and product names and does not imply Urdu OCR. Local category labels remain Must Have regardless of translation. Offline drafts do not require offline OCR or assistant processing; reminders do not automatically create recurring expenses.

5. Non-Functional Requirements

The following numerical targets are proposed FYP acceptance criteria from the detailed specification, not measured results. The team and supervisor should review them after an early pilot. Changes to a target should be explicit and justified; they must not silently remove the associated core capability. [2]

For performance and compatibility evaluation, use an Android 11-class device with 4 GB RAM and at least a 360 CSS-pixel-equivalent viewport, plus one newer Android version. Record the selected native or mobile-web deployment. The proposed network condition is at least 10 Mbps download, 2 Mbps upload and approximately 100 ms round-trip latency. Record actual device, build, provider configuration and cold or warm state.

The capacity fixture contains ten isolated test accounts with 1,000 active expenses each and up to ten items per receipt. Run a 20-minute mixed trial at one request per account every five seconds, with 20% writes and the remaining requests covering history, search and summaries. These conditions define an academic evaluation, not a production capacity promise.

ID	Quality requirement and verification	Priority
NFR-01	The system shall acknowledge ordinary actions within one second and display search/dashboard results within three seconds in at least 19 of 20 trials under the stated conditions. The capacity trial shall complete without lost or cross-account records; record latency, resource use and rejected requests.	Must Have
NFR-02	OCR and assistant requests shall show processing feedback within one second and a retryable timeout state by 60 seconds if unfinished. Measure visible waiting separately from provider latency; this is not a successful-completion guarantee.	Must Have
NFR-03	Credentials shall use no plaintext or reversible password storage and use salted adaptive hashing when managed locally. Proposed controls are a minimum 12-character password with support for at least 64, session expiry after 30 inactive minutes or 24 active hours, and throttling after five failed logins in 15 minutes or an equivalent non-enumerating control.	Must Have

ID	Quality requirement and verification	Priority
NFR-04	Remote financial communication shall use certificate-validated TLS. Stored records, images and local credentials shall have protected access; service secrets shall remain outside distributed clients and source control. Routine logs shall exclude raw financial content and tokens. Verify configuration and captured traffic/logs.	Must Have
NFR-05	The trusted data boundary shall reject all tested cross-account operations, including altered identifiers, exports and assistant requests. Test two accounts with overlapping merchant/product names, expired sessions and receipt text attempting to override retrieval restrictions.	Must Have
NFR-06	Receipt upload shall accept only verified JPEG or PNG images no larger than 10 MiB and reject invalid content before OCR. Test renamed executables, oversized files and malicious filenames alongside valid images.	Must Have
NFR-07	External processors shall receive only the receipt content or authorised financial subset needed for the requested operation. Explain processing before first use and inspect actual payloads and provider handling arrangements.	Must Have
NFR-08	Confirmed deletion shall remove an expense from active retrieval before success is reported and erase its live structured content and retained image within 24 hours. Protected backups containing the record shall expire within 30 days and remain inaccessible to the assistant. Verify storage and document provider retention.	Must Have
NFR-09	At least four of five representative trial users shall complete manual entry, receipt correction and transaction lookup without facilitator intervention after a brief introduction. Every specified invalid-input or service-error case shall identify the problem and a recovery action.	Must Have
NFR-10	OCR or assistant failure shall preserve already saved records and access through available core services, with user-triggered retry or manual recovery. Verify through service-failure tests; do not imply access without the connectivity required by the selected deployment.	Must Have
NFR-11	Interrupted saving shall leave either one complete saved expense or none. Repeating the same save request shall create no duplicates. Test interruptions and retries and inspect expense, item and allocation records together.	Must Have
NFR-12	PKR amounts shall be represented exactly to two decimal places; reviewed category allocations shall equal the expense total with zero difference after documented rounding. Journals shall balance wherever used. Verify mixed receipts and adjustments against known calculations.	Must Have

ID	Quality requirement and verification	Priority
NFR-13	The next fresh ledger, summary, price-history and assistant query shall reflect confirmed edits or deletions. Category/account/product corrections shall preserve valid references, distinct identities and original source text while the record exists. Verify these together across affected views.	Must Have
NFR-14	The assistant shall reproduce all expected numeric results in the controlled financial test suite and qualify or decline conclusions unsupported by authorised records. Include empty history, ambiguous periods, zero baselines, missing units, corrections and deleted transactions.	Must Have
NFR-15	Before user correction, extraction shall correctly propose at least 85% of legible payable totals and 70% of annotated purchase lines in the specified 20-receipt fixture. Record errors separately from successful manual correction and failure recovery.	Must Have
NFR-16	Core workflows shall complete on the stated Android device and one newer Android version. Entry, review, history and assistant controls shall remain usable without horizontal page scrolling at the stated viewport and 150% text scaling.	Must Have
NFR-17	An implemented Urdu release shall pass competent bilingual review of all translated core-screen strings, core assistant query types and mixed-direction layouts. Record language and presentation defects before claiming bilingual support.	Should Have
NFR-18	OCR, assistant retrieval and storage shall have documented replaceable interfaces and deterministic test substitutes. Each delivered requirement shall have implementation/test references, with affected integrity and access checks repeated after changes and optional deferrals recorded.	Must Have

The receipt fixture should contain 20 consented or synthetic printed receipts with three to twelve purchase lines, legible English or Latin-script text, varied layouts and known payable totals. Label each receipt's origin and reconcile two independent annotation passes. A correct purchase line includes its readable name and line total. Report merchant, date and quantity accuracy separately for fields actually present, with denominators. Keep blurred or cropped images separate for failure-path testing.

Repeat each controlled assistant query three times. Responses completed from an older record revision must be marked stale and refreshed. Security review should also document encryption-at-rest configuration, provider retention and training settings, and the proposed 15-minute login-throttle recovery interval. Invalid-input recovery should retain valid draft values and explain errors in text rather than relying only on codes or colour.

The prospective usability trial requires five users spanning the three primary groups. They have not yet been recruited or tested. Record consent, tasks, outcomes and difficulties. The controlled assistant suite verifies defined arithmetic and boundaries; it cannot demonstrate perfect answers to every possible question. Security tests similarly provide finite evidence and do not establish banking-grade certification.

6. Prioritisation

Priority	Business requirements	Functional requirements	Non-functional requirements
Must Have	6	22	17
Should Have	1	7	1
Could Have	1	4	0
Total	8	33	18

Implementation should first establish a complete capture-to-summary path with authentication, review, reliable saving and correction. Product classification, price history and the AI Financial Assistant must then complete the core product; their complexity does not make them optional. Existing OCR and language-model components may be integrated to keep the work feasible for a four-student team.

No Must Have function may depend on optional voice, Urdu localisation, exports, persistent offline drafts or extra screens. Shared quality controls apply to an optional feature when it is implemented. A dedicated month-comparison view can be deferred while the assistant still answers period comparisons. Optional image archiving can be deferred while structured receipt records remain usable.

7. Traceability Summary

The table links each D2 candidate to the consolidated business and functional requirements. D1 supplies the user needs behind those candidates. Shared non-functional conditions apply across relevant functions, particularly account isolation, coherent totals, failure recovery and correction consistency.

D2 requirement candidate	Business requirements	Functional requirements
DR-01 Receipt capture and review	BR-01, BR-02, BR-06	FR-OCR-01 to FR-OCR-03; FR-LED-01
DR-02 Manual and text cash entry	BR-01, BR-02	FR-CAP-01, FR-CAP-02; FR-LED-01
DR-03 Voice entry	BR-01	FR-OPT-01

D2 requirement candidate	Business requirements	Functional requirements
DR-04 Confirmable local categorisation	BR-02, BR-06	FR-CAT-01; FR-LED-01
DR-05 Accounting structure	BR-02	FR-CAT-02, FR-CAT-03
DR-06 Ledger and summaries	BR-02, BR-03	FR-LED-01, FR-LED-02, FR-LED-04
DR-07 Product classification and prices	BR-04	FR-PROD-01 to FR-PROD-03
DR-08 Specialised assistant	BR-03, BR-04, BR-05, BR-06	FR-AI-01 to FR-AI-04
DR-09 Account, privacy and maintenance	BR-02, BR-05, BR-06	FR-AUTH-01 to FR-AUTH-03; FR-LED-03
DR-10 Localisation, export and labels	BR-07	FR-OPT-02 to FR-OPT-04
DR-11 Visualisations, comparisons and archive	BR-07	FR-OPT-05 to FR-OPT-07
DR-12 Additional recording support	BR-08	FR-OPT-08 to FR-OPT-11

The links preserve the distinction between evidence and design. Receipt and cash capture draw strongly on reported recording friction. Accounting structure, semantic interpretation and assistant breadth additionally rely on the agreed project solution. Supporting one business outcome does not transfer its priority to every related optional feature: voice remains Should Have even though it contributes to BR-01.

8. Assumptions and Constraints

The implementation assumes access to legible supported receipts, users willing to review uncertain output, and affordable OCR and language-model components. Internet access may be necessary for selected services. When a service is unavailable, the application must communicate failure and support the applicable recovery path. Persistent offline recording remains optional.

Stored records may be incomplete relative to actual spending. Summaries and suggestions must therefore describe recorded data and relevant periods. Missing merchant, quantity or product information cannot be invented to make a record appear complete. User corrections take precedence over machine proposals, subject to valid totals and mappings.

The project is a four-student BS Computer Science FYP at Beaconhouse National University under Nouman Ali. It has no mandated vendor, programming language,

framework or database. Delivery mode, provider cost, deployment budget and retention arrangements require design decisions. Training a foundation model and operating infrastructure for millions of users are not commitments.

Providers must be evaluated for usable interfaces, data exposure, latency, cost and failure behaviour. Replaceable service interfaces and test substitutes support development when a provider is unavailable, but do not replace evaluation of the selected service. Representative devices, receipt annotations and trial participants must also be obtained before claiming empirical validation.

9. Out of Scope

The current project excludes the following:

- Direct bank integration, automatic debit-card or credit-card synchronisation, Easypaisa or JazzCash transaction integration, open banking and arbitrary third-party financial integrations.
- Payment processing, money transfers, commercial banking, investment management, tax filing, credit scoring and fraud-detection infrastructure.
- Sales, inventory, point-of-sale, payroll, enterprise ERP, complete SME accounting and enterprise multi-tenant organisation management.
- Shared household accounts, invitations, household permissions and collaborative editing in the current release.
- Banking-grade regulatory certification and production infrastructure for millions of concurrent users.
- Unrestricted financial-chatbot behaviour, investment advice, tax advice, banking advice and unrelated financial education.

An expense described as paid by a bank card or wallet remains a user-entered expense; it does not establish an integration. Similarly, internal account types do not create a business accounting interface, and a shopkeeper's supplier expense does not create a sales workflow.

10. Future Scope

Potential later work includes bank transaction importing, wallet integrations, shared household accounts, collaborative budgeting, business accounting features, inventory and point-of-sale functions. Other possibilities are recurring expense creation, budgeting goals, predictive analytics, anomaly detection, advanced financial recommendations, deeper Urdu support, additional languages, multi-currency support, retailer integrations and e-receipt ingestion.

These are possibilities beyond the current implementation commitment. Some are explicitly excluded now but may be reconsidered later. They must not be confused with current descriptive analysis: the assistant can analyse repeated past spending and unusually high recorded amounts, while automated recurring transactions and predictive

anomaly systems remain future work. Deeper Urdu support is also distinct from the current Should Have bilingual feature.

11. Validation and Open Issues

Validation should combine student review, supervisor feedback, controlled software tests and later representative-user evaluation. The current drafts specify what to check; they do not claim that the tests below have passed. Test records should identify the build, input, expected result, actual result and related requirement, with defects recorded when behaviour differs.

The following scenarios provide compact checks of the most important interactions:

1. **Mixed receipt:** process rice costing Rs. 100 and detergent costing Rs. 50, correct their categories and save. The draft contributes nothing; the saved expense contributes Rs. 150 overall, Rs. 100 to Ration and Rs. 50 to Household Supplies, never Rs. 300.
2. **Retrospective cash:** enter a Rs. 120 rickshaw fare for yesterday and another expense from last month. Confirm displayed dates and verify each is included in its purchase month rather than its creation month.
3. **Receipt adjustment:** review a Rs. 150 total whose item amounts sum to Rs. 160 because of a Rs. 10 discount. Saving remains blocked until allocations reconcile; the adjustment must not become a product-price observation.
4. **Conditional journals:** if double-entry is selected, record cash groceries of Rs. 2,500 with equal debit and credit amounts. Reject a journal with only Rs. 2,400 credited. If journals are not selected, verify expense allocations instead.
5. **Product comparison:** compare Surf Excel 1kg at Rs. 600 with Bonus 500g at Rs. 350. Retain separate identities and show Rs. 600/kg versus Rs. 700/kg. An item with unknown pack size receives no equivalent-unit conclusion.
6. **Assistant explanation:** compare Dining totals of Rs. 8,400 and Rs. 5,900. Report an increase of Rs. 2,500 with both periods. A suggestion to return to the earlier level must identify its hypothetical Rs. 2,500 reduction and assumptions.
7. **Correction and deletion:** change an expense's date, amount and category, then delete it. Verify consistent revisions and subsequent exclusion from fresh ledger, summary, product-history and assistant results, followed by the stated live-data removal.
8. **Privacy and failure:** try cross-account identifiers and receipt text asking the assistant to ignore access restrictions; disclose no protected records and perform no write. Interrupt saving and retry, producing one complete expense or none. Time out OCR and assistant requests and inspect recovery.

Additional checks should cover empty history, zero comparison baselines, invalid images, denied device permissions and selected optional features. Optional export should preserve values and neutralise formula-like CSV text where relevant. Urdu needs bilingual review,

and offline drafts need retry and duplicate checks. Deferred optional features should be recorded as deferred rather than presented as delivered.

Several issues remain open for review. The team should recover interview dates, consent context and primary records where possible, confirm Usman's participant age, and clarify the historical application-test context. These evidence gaps do not change the agreed persona definitions or create new research results.

The team and supervisor should review the proposed timing, capacity, extraction, usability, session and retention targets after a pilot. The delivery platform, account hierarchy, use of double-entry, providers, budget and backup/provider retention must be resolved during design. Selecting optional features, obtaining test devices and recruiting prospective participants are also outstanding. The D3 brief's conflicting mark heading and total should be clarified with the supervisor without affecting product requirements.

12. Conclusion

The consolidated catalogue preserves DailyKhata's expense-management scope while making it easier to review and develop. It connects D2 candidates to clear functions, retains measurable proposed quality conditions, and separates optional features from future work and exclusions. The next step is human review of these requirements and open issues, followed by design and implementation evaluation against the agreed behaviours.